



Linear regression

- Easy to understand.
- Foundational algo.

Linear Regression :

- Supervised ML (Regression Problem)

Simple LR

1 ip col.

1 op col.

Multiple LR

Multiple Ip col.

1 Op col.

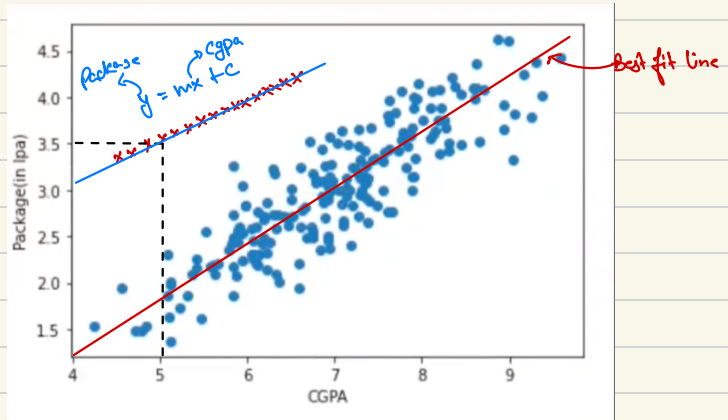
Polynomial LR

Regularization

Simple LR :

CGPA	Package.
7.1	3.5
4.7	1.2
8.9	4.2
8.1	3.9
⋮	⋮
⋮	⋮

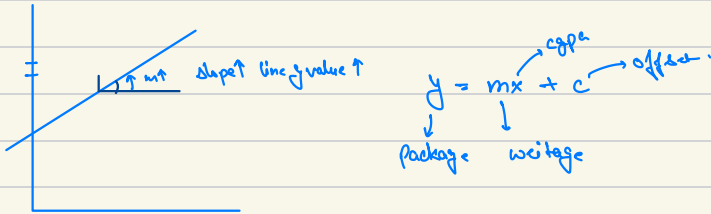
CGPA $\rightarrow$ Model $\rightarrow$ Package.



Not very linear.

Why? $\rightarrow$ Real world factors
(stochastic errors)

Intuition :



Offset : If $x=0$ then also y have some value.

eg.)

$x \rightarrow$ experience

package

$$y = mx + c$$

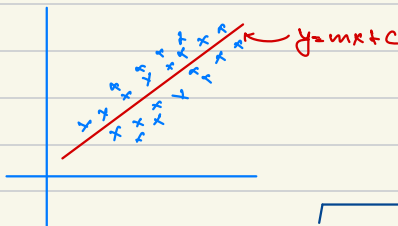
for fresher $x=0$

$$\therefore y = m \times 0 + c$$

$$y = c \leftarrow \text{offset}$$

as fresher will still have package.

How to find m and b ?



(m, b)

Closed form solution

Direct formula

OLS (Scikit Learn)

Ordinary Least Square.

Non-Closed form solution.

Gradient Descent.

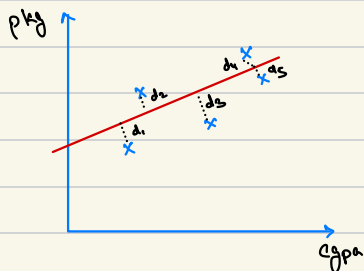
Used for higher dims.

SciD Regressor

OLS:

$$\bar{b} = \bar{y} - m\bar{x}$$

$$m = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$



$n \rightarrow$ total points.

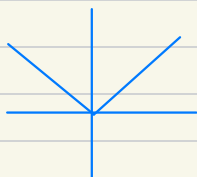
$$E = d_1 + d_2 + d_3 + \dots + d_n$$

$$E = d_1^2 + d_2^2 + d_3^2 + \dots + d_n^2$$

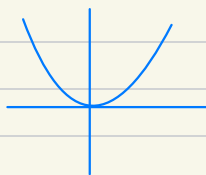
Why not $|d_1| + |d_2| + \dots + |d_n|$

Because we want to penalize outliers.

And we want to do differentiation.



Mod graph



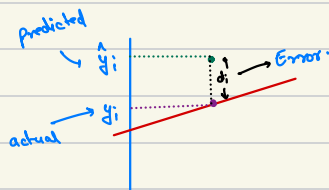
Sq. graph

$$E = \sum_{i=1}^n d_i^2$$

↑
Error fn. / Loss fn.

Also referred as J

$$E = \sum_{i=1}^n d_i^2$$



$$d_i = (y_i - \hat{y}_i)$$

$$E = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad \dots (1)$$

We need value of (m, b) that results the minimum E .

$$\hat{y}_i = mx_i + b \quad \text{substitution in (1)}$$

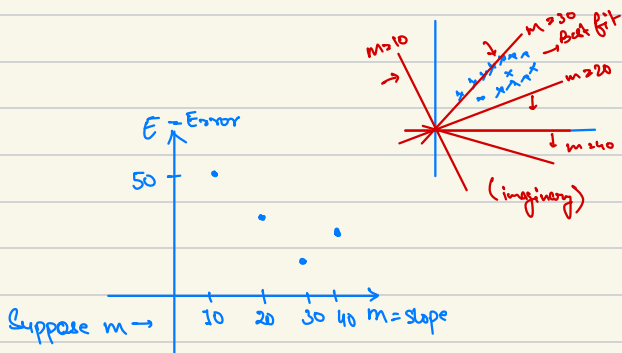
$$E(m, b) = \sum_{i=1}^n (y_i - mx_i - b)^2$$

$$E(m, b) = \sum_{i=1}^n (y_i - mx_i - b)^2$$

$$y = f(x)$$

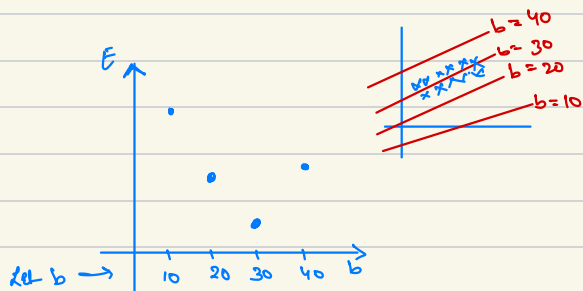
Let $b = 0$ (a constant)

$$E(m) = \sum_{i=1}^n (y_i - mx_i)^2$$



Let $m = 1$ (a constant)

$$E(b) = \sum_{i=1}^n (y_i - x_i - b)^2$$



Finding maxima & minima of E .

$$\frac{dE}{dx} = 0$$

$$\frac{\partial E}{\partial m} = 0, \quad \frac{\partial E}{\partial b} = 0$$

We get 2 eqn. at value of m, b .

Solving b :

$$\frac{\partial E}{\partial b} = \frac{\partial}{\partial b} \sum_{i=1}^n (y_i - mx_i - b)^2 = 0$$

$$\sum_{i=1}^n \frac{\partial}{\partial b} (y_i - mx_i - b)^2 = 0$$

$$\sum_{i=1}^n 2(y_i - mx_i - b)(-1) = 0$$

$$\sum_{i=1}^n \frac{-2(y_i - mx_i - b)}{-2} = \frac{0}{-2}$$

$$\sum_{i=1}^n y_i - mx_i - b = 0$$

$$\sum y_i - \sum mx_i - \sum b = 0$$

$$\frac{\sum y_i}{n} - \frac{\sum x_i}{n} - \frac{\sum b}{n} = \frac{0}{n}$$

$$\bar{y} - m\bar{x} - \frac{nb}{n} = 0$$

$$\bar{y} - m\bar{x} - b = 0$$

$$b = \bar{y} - m\bar{x} \quad \text{----- (1)}$$

Putting (1) in eqn of E.

$$E = \sum (y_i - mx_i - b)^2$$

↑
Putting (1).

$$E = \sum (y_i - mx_i - \bar{y} + m\bar{x})^2$$

$$\frac{\partial E}{\partial m} = \sum \frac{\partial}{\partial m} (y_i - mx_i - \bar{y} + m\bar{x})^2 = 0$$

$$\sum \frac{\partial}{\partial m} 2(y_i - mx_i - \bar{y} + m\bar{x})(-x_i + \bar{x}) = 0$$

$$\sum -2(y_i - mx_i - \bar{y} + m\bar{x})(x_i - \bar{x}) = 0$$

$$\sum (y_i - mx_i - \bar{y} + m\bar{x})(x_i - \bar{x}) = 0$$

$$\sum [(y_i - \bar{y}) - m(x_i - \bar{x})](x_i - \bar{x}) = 0$$

$$\sum [(y_i - \bar{y})(x_i - \bar{x}) - m(x_i - \bar{x})^2] = 0$$

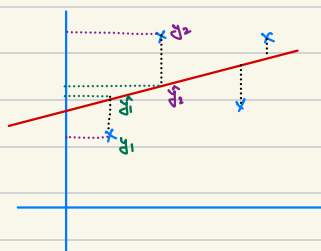
$$\sum (y_i - \bar{y})(x_i - \bar{x}) = m \sum (x_i - \bar{x})^2$$

$$m = \frac{\sum_{i=1}^n \frac{(x_i - \bar{x})(y_i - \bar{y})}{(x_i - \bar{x})^2}}{\dots\dots\dots} \quad \text{--- (2)}$$

Regression metrics :

- 1) MAE $\rightarrow$ Mean Absolute Error
- 2) MSE $\rightarrow$ Mean Squared Error
- 3) RMSE $\rightarrow$ Root Mean Squared Error
- 4) R^2 Score $\rightarrow$ Coefficient of determination.
- 5) Adjusted R^2 Score

MAE :



$$MAE = \frac{|y_1 - \hat{y}_1| + |y_2 - \hat{y}_2| + \dots + |y_n - \hat{y}_n|}{n}$$

$$MAE = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n}$$

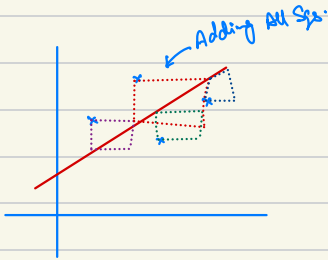
Advantage : • MAE is the no. $\rightarrow$ Loss

We want to reduce loss i.e MAE

- MAE is in units of y (same unit)
- Robust to outliers.

Disadvantage : • Graph of mod. fn is not differentiable at zero.

MSE :



$$MAE = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n}$$

$$MSE = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}$$

Advantage : • Can be used as loss fun.
as it's differentiable.

Disadvantage : • Difficult to interpret
the loss
• Penalize outliers
i.e. Not robust to outliers.

RMSE :

$$RMSE = \sqrt{MSE}$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}}$$

Advantage : • Same unit.

Disadvantage : • Not robust to outliers.

- Mostly used in DL.

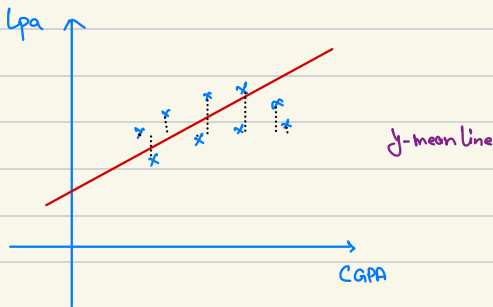
MSE, MAE → depends upon the context.

R² Score :

Tells how well model performs.

Independent of context.

CGPA	Package (in LPA)
------	------------------



R² Score : Coefficient of determination /
Goodness of fit

$$R^2 \text{ Score} : 1 - \frac{SS_R}{SS_M} \Rightarrow 1 - \frac{\left[\sum_{i=1}^n (y_i - \hat{y}_i)^2 \right]_{\text{Regression}}}{\left[\sum_{i=1}^n (y_i - \bar{y})^2 \right]_{\text{mean}}}$$

SS_R → Sum of Squared Errors in
Regression Line

SS_M → Sum of Squared Error in
Mean Line.

Compare mean line with regression line

R² Score : How better is regression line than mean line

If R^2 score = 0

then $\frac{SS_R}{SS_M} = 1$

i.e. The error done by regression line is equal to the error done by mean line.
 $SS_R = SS_M$

If R^2 score = 1

then $SS_R = 0$

i.e. Regression line is not doing any error; the regression line passes through every point, which is impossible.

$R^2 \rightarrow 1$: we want R^2 to go towards 1

$R^2 = -ve$: when $SS_R > SS_M$ i.e. regression line shows more error
(worst condition) than mean line.

Disadvantage :

R^2 Score CGPA | Lpa

→ On adding input cols, R^2 score increases.

i.e. It's assumed that more the columns, the variance of data is more explained.

CGPA | iq | Lpa
 ✓
 90%

→ If we add irrelevant col. (like temp), R^2 score increases or remains constant.

Adjusted R^2 Score :

$$\text{Adj } R^2 = 1 - \left[\frac{(1 - R^2)(n-1)}{(n-1-k)} \right]$$

$R^2 \rightarrow R^2$ Score

$n \rightarrow$ no. of rows.

$k \rightarrow$ total no. of independent cols. (i.e. input cols.)

Case 1: Adding temp col.

the value of k increases, so the term $(n-1-k)$ decreases.

hence, the fraction increases

$(1 - \text{fraction}) \rightarrow$ the entire term decreases.

Case 2: Adding relevant col. i.e. ig col.

$k \uparrow \rightarrow \therefore (n-1-k) \downarrow$

$R^2 \uparrow \therefore (1 - R^2) \downarrow$

$1 - R^2 \downarrow > (n-1-k) \downarrow$

fraction $\downarrow$

$(1 - \text{fraction}) \uparrow$

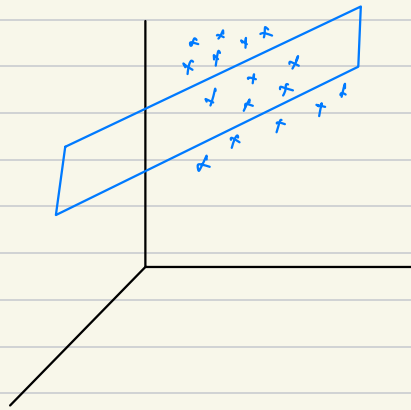
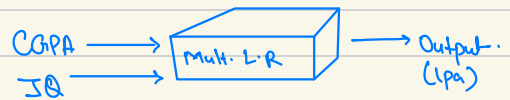
Multiple linear regression

Multiple Input Columns. (More than 1 predictors).



Multiple Linear Regression.

CGPA	IQ	Placement
8	80	8
9	90	9
5	120	15
⋮	⋮	⋮
1	⋮	⋮



$$y = \beta_0 x_1 + \beta_1 x_1 + \beta_2 x_2$$

$$x_1 \rightarrow \text{CGPA}$$

$$x_2 \rightarrow \text{IQ}$$

$$\beta_0 \rightarrow b$$

$$\beta_1 \rightarrow \text{w of } x_1$$

$$\beta_2 \rightarrow \text{w of } x_2$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

↑
hyper plane in n-dim co-ordinate.

$$(m.b) \quad \beta_0, \beta_1, \dots, \beta_n$$

↑
coefficient

Mathematical formulation :

	x_1	x_2	y
for	cgpa	iq	placement
	β_1	β_2	
$x_{11} \rightarrow$	8	$x_{12} \rightarrow$	80
$x_{21} \rightarrow$	7	$x_{22} \rightarrow$	70
$x_{31} \rightarrow$	5	$x_{32} \rightarrow$	120
			15

Let $\beta_0, \beta_1, \beta_2$ is known.

Prediction y for all students

$y_1 = 8$	$y_2 = 7$	$y_3 = 15$
$\hat{y}_1 = ?$	$\hat{y}_2 = ?$	$\hat{y}_3 = ?$

$$\hat{y}_1 = \beta_0 + \beta_1 \cdot 8 + \beta_2 \cdot 80$$

$$\hat{y}_2 = \beta_0 + \beta_1 \cdot 7 + \beta_2 \cdot 70$$

$$\hat{y}_3 = \beta_0 + \beta_1 \cdot 5 + \beta_2 \cdot 120$$

$$\hat{y}_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12}$$

$$\hat{y}_2 = \beta_0 + \beta_1 x_{21} + \beta_2 x_{22}$$

$$\hat{y}_3 = \beta_0 + \beta_1 x_{31} + \beta_2 x_{32}$$

Let there be m columns. and n rows.

$$\hat{y}_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \beta_3 x_{13} + \dots + \beta_m x_{1m}$$

$$\hat{y}_2 = \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \beta_3 x_{23} + \dots + \beta_m x_{2m}$$

$$\hat{y}_3 = \beta_0 + \beta_1 x_{31} + \beta_2 x_{32} + \beta_3 x_{33} + \dots + \beta_m x_{3m}$$

$$\vdots$$

$$\hat{y}_n = \beta_0 + \beta_1 x_{n1} + \beta_2 x_{n2} + \beta_3 x_{n3} + \dots + \beta_m x_{nm}$$

$$\hat{y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \hat{y}_3 \\ \vdots \\ \hat{y}_n \end{bmatrix}_{n \times 1} = \begin{bmatrix} \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \beta_3 x_{13} + \dots + \beta_m x_{1m} \\ \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \beta_3 x_{23} + \dots + \beta_m x_{2m} \\ \beta_0 + \beta_1 x_{31} + \beta_2 x_{32} + \beta_3 x_{33} + \dots + \beta_m x_{3m} \\ \vdots \\ \beta_0 + \beta_1 x_{n1} + \beta_2 x_{n2} + \beta_3 x_{n3} + \dots + \beta_m x_{nm} \end{bmatrix}_{n \times 1}$$

$$\hat{y} = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1m} \\ 1 & x_{21} & x_{22} & \dots & x_{2m} \\ 1 & x_{31} & x_{32} & \dots & x_{3m} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{nm} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_m \end{bmatrix}$$

$$\boxed{\hat{y} = X \cdot \beta} \quad \text{--- (eqn. 1)}$$

Diagram illustrating the dimensions of the variables in equation 1:

- $\hat{y}$ is of dimension $n \times 1$.
- X is of dimension $n \times (m+1)$.
- β is of dimension $(m+1) \times 1$.

Error fun for MLR.:

$$y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix}_{n \times 1} \quad \hat{y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \hat{y}_3 \\ \vdots \\ \hat{y}_n \end{bmatrix}_{n \times 1}$$

$$e = Y - \hat{Y} = \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix}_{n \times 1}$$

$$e^T \cdot e = \begin{bmatrix} y_1 - \hat{y}_1 & y_2 - \hat{y}_2 & \dots & y_n - \hat{y}_n \end{bmatrix}_{1 \times n} \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix}_{n \times 1}$$

$$e^T \cdot e = (y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + \dots + (y_n - \hat{y}_n)^2$$

$$= \sum (y_i - \hat{y}_i)^2$$

$$E = e^T \cdot e \quad \text{..... (eqn. 2)}$$

Minimizing Error :

$$E = (Y - \hat{Y})^T \cdot (Y - \hat{Y})$$

$$E = (Y^T - \hat{Y}^T) (Y - \hat{Y})$$

$$E = Y^T Y - \boxed{Y^T \hat{Y} - \hat{Y}^T Y} + \hat{Y}^T \hat{Y}$$

Same type of
matrix or
 $Y^T \hat{Y} = \hat{Y}^T Y$

$$Y^T \hat{Y} = \hat{Y}^T Y$$

$$\text{Let } A = Y, B = \hat{Y}$$

$$A^T B = \underline{B^T A}$$

$$(A^T B)^T \Rightarrow \underline{B^T A}$$

Now: $A^T B = (A^T B)^T$

$A^T B$ is symmetric

$$E = Y^T Y - 2Y^T \hat{Y} + \hat{Y}^T \hat{Y} \dots\dots\dots (\text{eqn 3})$$

Substituting (eq 1) in (eqn 3)

$$\hat{Y} = X\beta - g(1)$$

$$E = Y^T Y - 2Y^T X\beta + (X\beta)^T (X\beta)$$

$$E = Y^T Y - 2Y^T X\beta + X^T \beta^T X\beta \dots\dots\dots (\text{eqn 4})$$

$\uparrow$
 $E(\beta)$

$X \rightarrow \text{Data I/p} \quad Y \rightarrow \text{Data o/p.}$

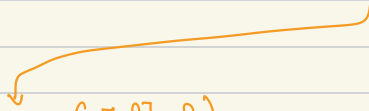
Now, we have to find such value of β matrix for which E is minimum.



$$\frac{dE}{d\beta} = 0$$

$$\frac{d}{d\beta} (Y^T Y - 2Y^T X\beta + X^T \beta^T X\beta) = 0$$

$$0 - 1 \cdot 2\gamma^T x + \frac{d(x^T B^T x B)}{d\beta} = 0$$



$$\frac{d(x^T B^T x B)}{d\beta}$$

Let

$$\begin{array}{ccccc} \beta^T & x^T & x & B \\ \downarrow & \downarrow & \downarrow & \\ x^T & A & x & \end{array}$$

$$\frac{d(x^T A x)}{dx} = 2x^T A$$

$$\therefore 2\beta^T x^T x$$

$$0 - 1 \cdot 2\gamma^T x + 2\beta^T x^T x = 0$$

$$2\beta^T x^T x = 2\gamma^T x$$

Now, finding the value of β :

Multiplying $(x^T x)^{-1}$ both side

$$2\beta^T x^T x (x^T x)^{-1} = 2\gamma^T x (x^T x)^{-1}$$

$$\beta^T I = \gamma^T x (x^T x)^{-1}$$

$$\beta^T = \gamma^T x (x^T x)^{-1}$$

Transposing both side.

$$(\beta^T)^T = \left(\underbrace{\gamma^T}_\text{Let } A} \underbrace{x(x^T x)^{-1}}_B \right)^T$$

$$\alpha = x^T A x$$

$$\frac{d\alpha}{dx} = 2x^T A$$

A should be symmetric

$$\text{Symmetric: } (A^T)^T = A$$

$$A^{-1} \cdot A = I$$

$$A \cdot I = A$$

$$(\beta^T)^T = [B \ A]^T$$

$$\beta = [(X^T X)^{-1}]^T (Y^T X)^T$$

$$\beta = [(X^T X)^{-1}]^T X^T Y$$

Proving $(X^T X)^{-1}$ is symmetric.

$$\text{Let } A = (X^T X)^{-1} \longrightarrow A^T \cdot A = A$$

$$A \cdot A^T = I.$$

$$(A \cdot A^T)^T = I^T$$

$$(A^T)^T A^T = I$$

$$(A^{-1})^T A = I$$

$$\text{Mult. } A^{-1}$$

$$(A^{-1})^T A A^{-1} = I A^{-1}$$

$$(A^{-1})^T I = A^{-1}$$

$$(A^{-1})^T = A^{-1}$$

OLS
(Ordinary Least Square)
formula.

$$\beta = (X^T X)^{-1} X^T Y \quad \text{..... (eqn 5.)}$$

$$\begin{array}{ccc} (X^T X)^{-1} & X^T & Y \\ [(m+1) \times n \quad \underbrace{n \times (m+1)}] & (m+1) \times n & n \times 1 \end{array}$$

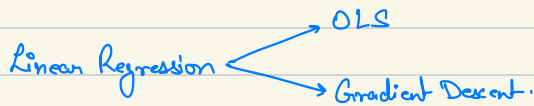
$$\begin{array}{ccc} [(m+1) & (m+1)] & (m+1) \times n \quad n \times 1 \\ \underbrace{\hspace{10em}} & & \end{array}$$

$$\begin{array}{ccc} (m+1) \times n & & n \times 1 \\ \underbrace{\hspace{10em}} & & \end{array}$$

$$(m+1) \times 1$$

$$\text{Shape of } \beta = (m+1) \times 1$$

Problems with OLS :



In OLS (Ordinary Least-Squared) we want to find inverse.

$$\beta = (x^T x)^{-1} x^T y$$

↑ inverse

which is computationally expensive.

$$[\text{Time Complexity} = O(n^3)]$$

$$\begin{array}{cc} x^T & x \\ (m+1) \times n & n \times (m+1) \\ \hline & (m+1)(m+1) \\ & (\text{ilp.col} + 1)(\text{ilp.col} + 1) \end{array}$$

Matrix shape →

Let ilp.col = 9
then (10×10)
 (100)

$$\begin{aligned} \text{P.C.} &= O(n^3) = O(100)^3 \\ \text{for inverse} &= O(1000000) \end{aligned}$$

for high dimension data.

$$100 \times 100$$

$$\text{then } O(10000)^3$$

